

Ibrahim Kecoglu

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EDUCATION

PhD, Biomedical Physics at **Stanford University** 2024 - 2030
BS, Mathematics at **Bogazici University** (High Honors, GPA: 3.79/4.00) 2022 - 2024
BS, Physics at **Bogazici University** (High Honors, 2nd Place, GPA: 3.79/4.00) 2020 - 2024

RESEARCH EXPERIENCE

Durmus Lab – Stanford University (Supervisor: Gozde Durmus)
Graduate Researcher

June 2025 – Present

- Built a magnetic levitation Raman spectroscopy platform to sort and characterize microliter-scale biological samples.

PhD Lab Rotation

Sep. 2024 – Dec. 2024

- Developed computational simulations to study cell aggregation in oscillating-field magnetic levitation devices.
- Optimized SERS substrate performance by evaluating surface treatment protocols.
- Applied SERS for exosome detection, improving the sensitivity of nanoscale vesicle analysis.
- Gained hands-on wet lab experience, including cell culture, biosafety cabinet practices, and quantitative cell assays.

Loo Lab – Stanford University (Supervisor: Billy W. Loo)
PhD Lab Rotation

March 2025 - June 2025

- Designed and constructed a lateral beam irradiation setup for mouse brain studies.

Molecular Imaging Instrumentation Laboratory – Stanford University (Supervisor: Craig S. Levin)

PhD Lab Rotation

Jan. 2025 – Mar. 2025

- Developed a generative deep learning model to synthetically enhance PET images, producing higher-resolution outputs from conventional PET data.

Berker Research Group – MIT & Kadir Has University (Supervisor: Ahmet Nihat Berker)
Undergraduate Researcher

Mar. 2020 – June 2024

- Investigated statistical mechanics models such as Ising, Potts, and Ashkin–Teller.
- Characterized phase diagrams and transitions.
- Applied advanced methods such as renormalization group theory, duality transformations, transfer matrix methods, Migdal–Kadanoff approximation, and multifractal analysis.
- Developed a novel framework for classifying chaotic behavior in real-world systems.
- Authored three peer-reviewed publications (two first-author, one co-author).

Medical and Biological Physics Laboratory – Bogazici University (Supervisor: Mehmet Burcin Unlu)

Undergraduate Researcher

Aug. 2019 – June 2024

- Designed and built scanning Raman spectroscopy, optical tweezer, and microscopy setups.
- Conducted experiments on diverse biological samples, including animal tissues, blood, plant tissues, and nano-scale vesicles (exosomes).
- Applied spectral analysis methods (e.g., band component analysis) and built statistical and machine learning models in MATLAB and Python.
- Performed biological sample preparation, including extracellular vesicle isolation from cell media.
- Authored seven peer-reviewed publications (one first-author, six co-author).

TEACHING EXPERIENCE

Stanford University - Teaching Assistant

Sep. 2025 - Dec. 2025

- BMP/RAD/BIOE 222: Physics and Engineering Principles of Multi-modality Molecular Imaging of Living Subjects

KHAS University Summer/Winter School - Teaching Assistant

July 2019 - June 2024

- Phase Transitions and Renormalization-Group Theory
- Electricity and Magnetism
- Introduction to Quantum Mechanics
- Introduction to Computer Engineering

AWARDS & SCHOLARSHIPS

TUBITAK 2205 - Undergraduate Scholarship Program

Sep. 2019 - June 2024

Awarded to students in the top 1% nationwide pursuing STEM studies.

EpiSTEM Research Competition - 1st Place

Apr. 2019

Designed and built a DIY spectrometer and investigated the effect of the storage surface on the decomposition of fruits.

LICENSES & CERTIFICATIONS

Introduction to Molecular Biology and Genetics Engineering at Kadir Has University	2020
Social and Physical Networks, Complex Systems, Chaos and Control at Kadir Has University	2020
Phase Transitions and Renormalization Group at Kadir Has University	2020
Big Data - Intelligent Data: You Too Can Create It, Manage It at Kadir Has University	2019
Introduction to Computer Engineering at Kadir Has University	2019
Introduction to Quantum Mechanics at Kadir Has University	2018

SKILLS

Technical Skills

- **Programming & Data Analysis:** Python (NumPy, PyTorch, scikit-learn, SymPy), MATLAB, COMSOL, Mathematica, spectral data analysis, instrument control
- **Machine Learning & AI:** CNNs, U-Net, GANs, diffusion models, Random Forests, dimensionality reduction (PCA, t-SNE)
- **Experimental Techniques:** Photonics, Raman spectroscopy, optical tweezers, magnetic levitation, PET/CT imaging
- **Laboratory Skills:** Biological sample preparation, extracellular vesicle isolation, cell culture, SERS substrate preparation, microscopy, biosafety cabinet work
- **Theoretical & Computational Physics:** Statistical mechanics, renormalization-group theory, duality transformations, transfer matrix methods, multifractal analysis, spin-glass theory

Language Skills

- Turkish: Native
- English: Advanced academic proficiency (10 peer-reviewed publications in English)

PUBLICATIONS

- [10] Arda Inanc, Nayce Ilayda Bektas, **Ibrahim Kecoglu**, Ugur Parlatan, Begum Durkut, Melike Ucak, Mehmet Burcin Unlu, and Ciler Celik-Ozenci. “Label-free differentiation of functional zones in mature mouse placenta using micro-Raman imaging”. In: *Biomedical Optics Express* 15.5 (Apr. 2024), pp. 3441–3456. ISSN: 2156-7085. DOI: [10.1364/BOE.521500](https://doi.org/10.1364/BOE.521500).
- [9] **Ibrahim Kecoglu** and A. Nihat Berker. “Global Ashkin–Teller phase diagrams in two and three dimensions: Multicritical bifurcation versus double tricriticality—endpoint”. In: *Physica A: Statistical Mechanics and its Applications* 630 (Nov. 2023), p. 129248. ISSN: 0378-4371. DOI: [10.1016/j.physa.2023.129248](https://doi.org/10.1016/j.physa.2023.129248).
- [8] E. Can Artun, **Ibrahim Kecoglu**, Alpar Türkoğlu, and A. Nihat Berker. “Multifractal spin-glass chaos projection and interrelation of multicultural music and brain signals”. In: *Chaos, Solitons & Fractals* 167 (Feb. 2023), p. 113005. ISSN: 0960-0779. DOI: [10.1016/j.chaos.2022.113005](https://doi.org/10.1016/j.chaos.2022.113005).
- [7] Ayse SEN, **Ibrahim Kecoglu**, Muhammad Ahmed, Ugur Parlatan, and Mehmet Burcin Unlu. “Differentiation of advanced generation mutant wheat lines: Conventional techniques versus Raman spectroscopy”. In: *Frontiers in Plant Science* 14 (Feb. 2023). ISSN: 1664-462X. DOI: [10.3389/fpls.2023.1116876](https://doi.org/10.3389/fpls.2023.1116876).
- [6] Ugur Parlatan, Mehmet Ozgun Ozen, **Ibrahim Kecoglu**, Batuhan Koyuncu, Hulya Torun, Davod Khalafkhany, Irem Loc, Mehmet Giray Ogut, Fatih Inci, Demir Akin, Ihsan Solaroglu, Nesrin Ozoren, Mehmet Burcin Unlu, and Utkan Demirci. “Label-Free Identification of Exosomes using Raman Spectroscopy and Machine Learning”. In: *Small* 19.9 (Jan. 2023), p. 2205519. ISSN: 1613-6829. DOI: [10.1002/smll.202205519](https://doi.org/10.1002/smll.202205519).
- [5] Irem Loc, **Ibrahim Kecoglu**, Mehmet Burcin Unlu, and Ugur Parlatan. “Denoising Raman spectra using fully convolutional encoder–decoder network”. In: *Journal of Raman Spectroscopy* 53.8 (May 2022), pp. 1445–1452. ISSN: 0377-0486, 1097-4555. DOI: [10.1002/jrs.6379](https://doi.org/10.1002/jrs.6379).
- [4] **Ibrahim Kecoglu**, Merve Sirkeci, Mehmet Burcin Unlu, Ayse Sen, Ugur Parlatan, and Feyza Guzelcimen. “Quantification of salt stress in wheat leaves by Raman spectroscopy and machine learning”. In: *Scientific Reports* 12.1 (May 2022), p. 7197. ISSN: 2045-2322. DOI: [10.1038/s41598-022-10767-y](https://doi.org/10.1038/s41598-022-10767-y).

- [3] Ugur Parlatan, Seyma Parlatan, Kubra Sen, **Ibrahim Kecoglu**, Mustafa Ozer Ulukan, Atalay Karakaya, Korhan Erkanli, Halil Turkoglu, Murat Ugurlucan, Mehmet Burcin Unlu, and Bukem Tanoren. “Atrial fibrillation designation with micro-Raman spectroscopy and scanning acoustic microscope”. In: *Scientific Reports* 12.1 (Apr. 2022), p. 6461. ISSN: 2045-2322. DOI: [10.1038/s41598-022-10380-z](https://doi.org/10.1038/s41598-022-10380-z).
- [2] Bukem Tanoren, Ugur Parlatan, Melita Parlak, **Ibrahim Kecoglu**, Mehmet Burcin Unlu, Didem Melis Oztas, Mustafa Ozer Ulukan, Korhan Erkanli, and Murat Ugurlucan. “Aortic aneurysm evaluation by scanning acoustic microscopy and Raman spectroscopy”. In: *Analytical Methods* 13.39 (Sept. 2021), pp. 4683–4690. ISSN: 1759-9660, 1759-9679. DOI: [10.1039/D1AY01133B](https://doi.org/10.1039/D1AY01133B).
- [1] **Ibrahim Kecoglu** and A. Nihat Berker. “Across dimensions: Two- and three-dimensional phase transitions from the iterative renormalization-group theory of chains”. In: *Physical Review E* 102.3 (Sept. 2020), p. 032134. ISSN: 2470-0045, 2470-0053. DOI: [10.1103/PhysRevE.102.032134](https://doi.org/10.1103/PhysRevE.102.032134).